

1. CTBI Sanity Check (on Dev's real CTBI.csv, 6 heatwave seasons, 2021–2026)

Ran 4 independent behavioural checks per season, not a re-derivation of the formula:

Check	What it verifies	Result
Rises to peak	CTBI increases from season start to seasonal peak	✓ 6/6 seasons
Declines after peak	CTBI falls again after peak (recovery behaviour)	✓ 6/6 seasons
Near-zero early NCTL	First 5 days of April show ~0 nighttime load	✓ 6/6 seasons
NCTL↔CTBI correlation	A season's max NCTL correlates with its peak CTBI	✓ $r = 0.944$

Verdict: CTBI behaves exactly as physically expected across all 6 seasons. Full detail in docs/Khushi_Day3_CTBI_Sanity_Check.md.

2. Priority Rubric v1 — built and documented

Rule-based, transparent, not ML (per playbook Day 4 spec, built one day early since CTBI was ready):

Full rule (once vulnerability + response capacity are real):
High thermal + High vulnerability + Low response capacity → CRITICAL
High thermal + (High vulnerability OR Low response capacity) → HIGH
Moderate thermal + High vulnerability → MODERATE
Everything else → LOW

Thermal tier thresholds are derived from the **real CTBI distribution** (33rd/66th percentile of Dev's 546-day series): Low < 0.382 ≤ Moderate < 0.443 ≤ High.

3. Tested on real ward-level CTBI (Garima's ward_thermal_daily.csv)

- Ran on all 48 wards × 546 days (26,208 rows) —
data/processed/priority_rubric_v1_full_test.csv

- Ran on latest-date snapshot —
data/processed/priority_rubric_v1_test_snapshot.csv
- **Monthly thermal-tier distribution behaves sensibly:** April is mostly Low (66%), June is mostly High (54%) — matches Ahmedabad's actual heatwave escalation pattern (April mild → June peak).

Known limitation (by design, not a bug)

Because **vulnerability and response-capacity ward data are still blocked** (2011 Census ward-delimitation mismatch — 57 old wards vs 48 current wards, discussed and decided not to force-fit), the rubric currently returns:

```
"Thermal-only: <tier> (partial – vulnerability/response capacity pending)"
```

instead of a real CRITICAL/HIGH/MODERATE/LOW verdict. **This is intentional** — the rubric never claims a vulnerability-based verdict it can't yet justify, per the project's honesty rule (no fabricated demographic data → no fabricated priority levels either).

Also expected: on any single date, all 48 wards currently show the *same* CTBI value. This is because ward_thermal_daily.csv attaches one city-wide CTBI value per date to every ward (CTBI.csv itself is a single time series, not per-grid-point). This is already a known/documented Day 3 limitation, not new.

What unblocks the full rubric

Once NDVI (green cover, via Sentinel-2) and healthcare-access (via OSM hospital distance) are computed per ward — the plan agreed on to replace the blocked Census data — this same priority_rubric_v1() function plugs them in directly (just pass vulnerability= and response_capacity= instead of None) and starts returning real CRITICAL/HIGH/MODERATE/LOW without any code changes.